Live app: https://asme326si3qdbv4raxgm4fbbge0fohve.lambda-url.us-east-1.on.aws/ · Code: https://github.com/woooj02/woundtrace · Video: https://youtu.be/AAn2KQZTjbw

WoundTrace: technical report

OpenCV AI Competition 2026 · solo entry by aryan-neopane · live: https://asme326si3qdbv4raxgm4fbbge0fohve.lambda-url.us-east-1.on.aws/

1. Problem and users

Chronic wounds are common and growing: Medicare beneficiaries with a wound rose from 8.2 million in 2014 to 10.5 million in 2019 (Carter et al. 2023; the two years used different Medicare data sets) [3]. Between visits, home-health and wound-clinic nurses track wounds by area. Two numbers matter: how big the wound is, and how much it has shrunk. For diabetic foot ulcers, the Society for Vascular Surgery recommends considering adjunctive therapy when the area has not fallen by more than 50% after at least 4 weeks of standard care [1]; the 4-week change predicts 12-week healing (Sheehan et al. 2003, n = 203) [2].

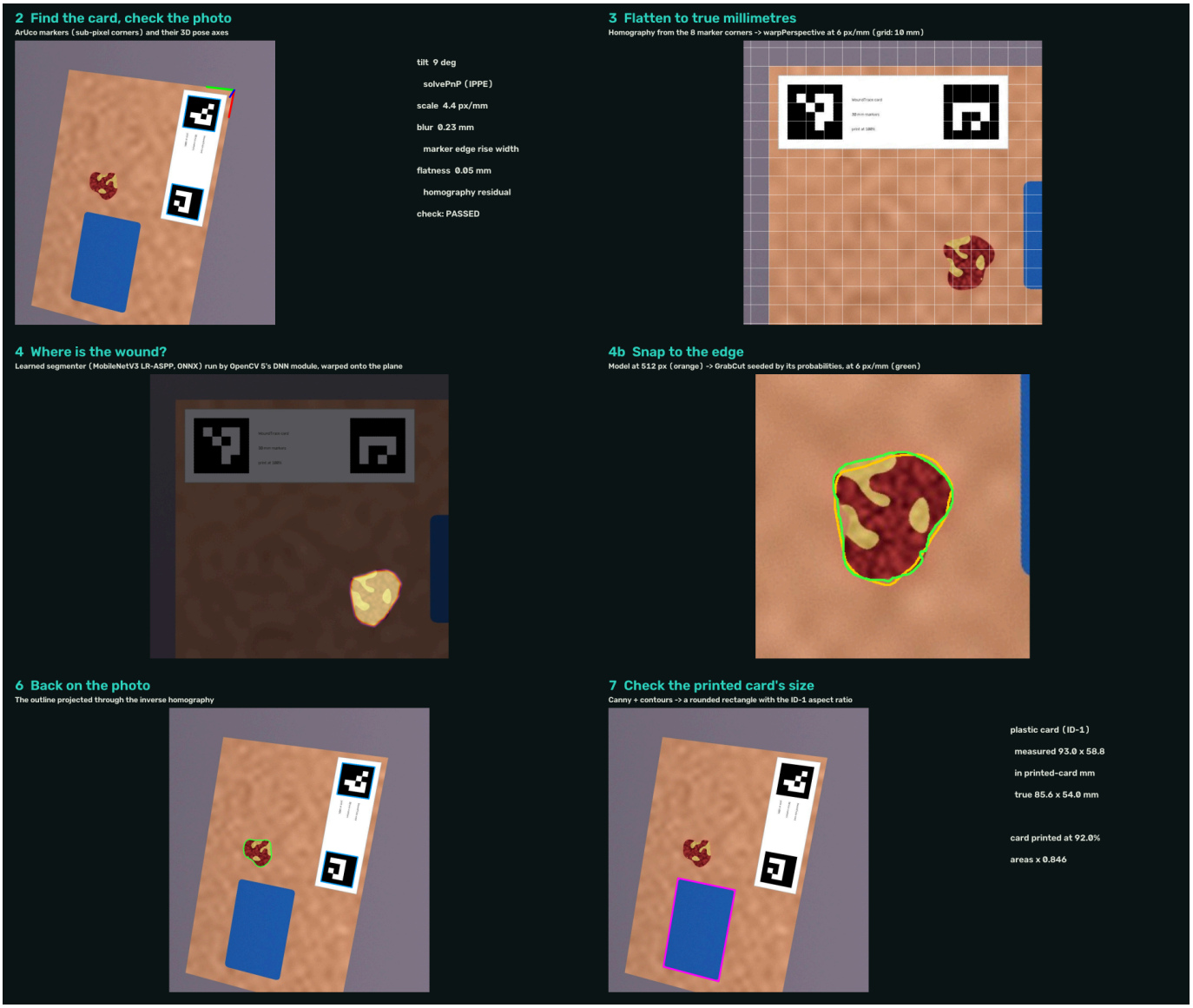
Measuring at home fails in three ways: a ruler's length × width overestimates irregular wounds; phone photos are tilted, blurred or lit by a flash; and a card printed at home is often printed at the wrong size. **User:** a nurse following patients who send photos from home. **Promise:** one photo next to a printed card becomes an area in cm^2 with its uncertainty, or a specific instruction to take a better photo; stalled healing becomes a drafted note that the nurse approves.

2. What it does

One visit (diagram: docs/architecture.svg):

1. `check_photo`: card, tilt, blur, resolution, flatness, brightness. Fail → `request_retake` with the fix (no card: the DNN and the vision model still show where the wound is, with no size).
2. `calibrate_card` (once per printed card): measure the printed card's true size from a plastic bank card.
3. `measure_wound`: rectify to true millimetres, outline the wound, measure area ± uncertainty, length, width, perimeter; the clinic's rules decide record, retake (glare), measure again with a second method (weak edge), or ask the nurse to confirm the outline.
4. `review_outline`: the vision model marks the wound on the plain photo; if its box and OpenCV's outline disagree, the outline goes to the nurse instead of the record.
5. `get_history`, `assess_healing`: growth counts as worsening only if it exceeds 10% **and** both photos' noise.
6. `record_measurement`; `request_nurse_review` drafts a note that waits until a human approves it.

3. OpenCV 5 pipeline (`app/vision.py`)



- **Card:** two 30 mm ArUco markers (`DICT_4X4_50`), `ArucoDetector` with sub-pixel corner refinement. Eight corners → `findHomography` to card millimetres; its RMS residual (mm) checks the card is flat.
- **Quality:** tilt from `solvePnP` (IPPE, assumed phone focal length); blur as the 10-90% rise width of the markers' outer edges, sampled along normals with `remap`, in mm; resolution in px/mm; brightness of the card's white strip. Gates: ≥ 2.5 px/mm, tilt ≤ 45°, edge ≤ 1.3 mm, residual ≤ 0.6 mm, brightness ≥ 70.
- **Rectification:** `warpPerspective` onto the skin plane at 6 px/mm.
- **Segmentation (default):** a learned segmenter (LR-ASPP, MobileNetV3-Large; model card) exported to ONNX and run by `cv2.dnn` on the whole photo at 512 px; no PyTorch at runtime. Its probability map is warped onto the plane with the same homography, and `grabCut` seeded from it (sure > 0.9, likely

  > 0.5, background beyond 8 mm) snaps the outline to the colour edge at 6 px/mm. The most confident component that is off the card and not cut by the frame is the wound. Trained on the FUSeg and CO2Wounds-V2 train splits, 1,242 healthy-feet photos as no-wound examples and 500 synthetic photos (`eval/train_segmenter.py --more-data`, ~45 min on a GTX 1660 Ti).

- **Segmentation (classical, the cross-check):** Lab distance from the skin colour, a background-free Otsu split, a ring skin sample, an edge-seeking threshold (strongest Sobel boundary of 16), slough

handling, then `grabCut`. Line-like or thin shapes (aspect > 8 or nothing ≥ 3 mm across, from `minAreaRect` and `distanceTransform`) are not wounds.
- **Measurement:** area, Feret length and width, the ruler's L × W, perimeter, solidity, glare, edge contrast.
- **Card calibration:** rectify a wide window, `Canny` + `findContours`, keep rounded rectangles with the ID-1 aspect ratio (85.60 × 53.98 mm); average the inner and outer contour of the edge line; print scale = true size / measured size, both sides must agree within 2.5%. The plastic card is masked out of segmentation.
- **Uncertainty:** relative $sd^2 = k0^2 + (k1 \cdot \text{perimeter} \cdot (\text{edge blur} + \text{pixel size}) / \text{area})^2 + (k2 \cdot \text{card residual})^2$, fitted by maximum likelihood on the tuning seed (k0 = 0.0039, k1 = 0.205, k2 = 0).

4. Agent (`app/agent.py`, `app/tools.py`)

Amazon Bedrock Converse with tool use (Nova 2 Lite), up to 10 steps, 9 tools. The model chooses tools and words messages; plain code enforces the rules: no measuring a failed photo, no recording without a "record" decision or an unchecked card, no retake for a measurable photo, ids pinned to the visit, nurse notes queued for approval. **Outline check** (`app/review.py`): the same model is shown the plain photo (never OpenCV's outline) and asked for the wound's bounding box; code computes the IoU with the outline's box and a value below 0.1 sends the outline to the nurse. A Bedrock visit cannot record before this check. An **outgoing-message guard** checks each patient reply for leaked internals, clinical opinions ("looks good") and length, and substitutes the template message if it fails. A deterministic rules engine runs the same tools when Bedrock is unavailable or the daily cap (150 visits) is reached; the UI always says which engine ran. Each visit is stored as a trace in S3 and logged as one JSON line to CloudWatch.

5. Evaluation

Real wounds: outline (`eval/fuseg_summary.md`, `eval/external_summary.md`). Three public datasets of de-identified photos with annotators' masks, none with a card, so only the outline is tested. Only held-out photos are scored. "v1" is the first learned model (FUSeg train split only); "v2" is the deployed one.

Held-out set	Photos	Classical	v1	v2 (deployed)
FUSeg 2021 validation (US clinic, foot ulcers)	200	0.03 (97 missed)	0.84 (2 missed)	**0.82** (5 missed)
CO2Wounds-V2 validation (Colombia, leprosy, smartphones)	122	–	0.48 (14 missed)	**0.77** (2 missed)
Lower-limb set, never trained on (several wound types)	2,686	–	0.33 (378 missed)	**0.39** (428 missed)
Healthy feet, held-out half: given an outline	1,378	–	23%	**0%**

Mean Dice, with the GrabCut snap. On the lower-limb set v2's outlines sit on the wound (precision 0.83) but cover less than its masks, which include surrounding skin (recall 0.37). Estimated skin tone (ITA° of the skin around the wound) made little difference: Dice 0.39-0.46 in each of six bins (n = 221-978). The 0% on healthy feet may partly reflect that source's look: the no-wound training photos came from it too. Examples (CC BY 4.0 set only): docs/real_examples.png.

Outline check (`eval/review_summary.md`). 261 outlines on the FUSeg photos (v1 and classical, so both good and bad ones), excluding the 40 used to design the check. Good: Dice ≥ 0.7; bad: < 0.4.

Outlines	Passed	Sent to the nurse
Good (159)	72%	28%
Bad (86)	21%	**79%**

Synthetic photos with known area. `app/synth.py` draws an irregular wound (granulation, slough, periwound redness) on one of six skin tones and photographs it with a simulated camera (perspective, distance, defocus, flash glare, noise, JPEG); the true area is exact. Seed 2026 tuned the gates and the uncertainty; seed 7 was used to diagnose failures; **seed 11 is held out** (600 hard + 150 clean photos).

Synthetic, held-out	Result
Recorded / retake / nurse outline (hard set)	245 / 355 / 0
Mean area error when recorded · 95th percentile	1.5% · 5.5% (v1: 1.2% · 4.4%)
Recorded with error > 10%	3 (v1: 1)
Same pipeline with gates off · clean set	5.8% · 0.7% mean error
Ruler L × W vs true area	+43.3%
Error by skin tone 1 → 6 (light → dark)	2.4 · 1.1 · 1.2 · 1.2 · 1.3 · 1.6%
True area inside ± 1 sd / ± 2 sd	79.9% / 95.6% (targets 68% / 95%)

Real phone photos with known area (26 iPhone photos, room light; `real/summary.md`). Not wounds: colour sheets A and B from a faulty printer (banding), a black-ink sheet, and a $1 coin (25.00 mm). Card found 26/26. v2 recorded 2 (−3.5% and −7.5%), sent 17 to the nurse to confirm the outline (11 of them within 10%, sheet B up to −27%), and asked for 7 retakes: 2 too far, 1 bent card, and 4 "no wound" (one sheet B photo, the black-ink sheet and both coin photos; v1 had measured the last three at +1.7%, +5.7%, +6.1%). Only 1 of the 2 recorded fell inside ± 2 sd.

Card calibration (`eval/calibration_summary.md`). In 12 synthetic photos with cards printed at 92.5-104% and tilts of 5-35°, the plastic card was found in 11 (missed at 35°, which leads to a retake); corrected areas were within 2.6% of the truth (uncorrected: up to +18%).

Agent on Bedrock (`eval/agent_summary.md`): 60 visits, 10 scenarios × 6 skin tones, each also run through the rules engine from the same state.

Result	Value
Same outcome as the rules (record / retake / which nurse request)	60/60
Outline check on the 29 visits that measured a wound	agreed 29/29 (synthetic photos)
Tool calls refused by the guards	0
Numbers in patient messages that came from tool outputs	100%
Replies replaced by the message guard	2
Latency per visit, median / p95	5.3 / 8.9 s
Tokens per visit, input / output	16,363 / 241 (≈ $0.006 at third-party listed Nova 2 Lite prices [5])

The agent run used the card printed at 92.5%: measured scale 0.921, recorded area error 1.3%.

6. Failure cases found, and what changed

1. **The classical pipeline failed on real wounds.** Tuned on synthetic targets, it scored 1.4% area error there, then Dice 0.03 on real foot ulcers, missing 97 of 200: real wound beds, callus, toes and bedding break a colour rule. Replaced as the default by the learned segmenter (Dice 0.84); kept as a cross-check.
2. **The learned model did not travel.** On two other datasets v1 scored 0.48 and 0.33 and drew outlines on 23% of healthy feet. Retrained with CO2Wounds' train split and healthy feet as no-wound examples (v2): 0.77 and 0.39, no false outlines, at a cost of 0.02 Dice on FUSeg and 0.3 points of synthetic area error.
3. **The outline check passed everything.** First design: show the model the photo and the outline overlay and ask whether it follows the wound. It answered "follows" for 100% of good **and** 100% of bad outlines (`eval/review_overlay_summary.md`). Redesigned so the model never sees the outline: it marks the wound itself and code compares. Now 79% of bad outlines are caught, at the cost of 28% false alarms.
4. **Dark skin, slough only** (classical, seed 7): 7 photos recorded 64-98% too small on skin tones 4-6. Fixed with a background-free re-split, a ring skin sample and a deep-core rule.
5. **Printed text measured as a wound** (real sheet A): line and thickness filters plus a policy backstop.
6. **Misprinted card.** The coin read +24.5% because the printer shrank the card to 92.3% (confirmed with a ruler); the photo looks perfect. Added card calibration; nothing is recorded until the card is checked.
7. **Agent reasoning sent to patients.** In the first agent run, 16/60 retake replies were the model's working notes and 2 said "your wound looks good". Added the message guard and use the model's own `request_retake` message.
8. **Conservative gates.** 208 of 323 measurable retakes would have measured within 5%: the gates trade patient effort for few bad accepts. Loosening them needs real-wound data with known area.

7. Limitations

- **Area accuracy on real wounds is untested.** The real-wound tests are outlines on clinic photos without a card; patients' own photos, depth and curved limbs are untested. Skin tone is estimated from the photos, not recorded by the datasets.
- On a clinic it never trained on, the model misses 16% of wounds and outlines less than annotators mark.
- The outline check sends 28% of good outlines to the nurse; the ± range is too narrow on real photos.
- Area only (no depth or tissue types). Tilt assumes a focal length; the card must lie in the wound's plane.
- The healing threshold comes from diabetic foot ulcer evidence; for other wounds it is a configurable prompt.

8. Responsible use

A measurement aid, never a diagnosis, outcome prediction or treatment recommendation. Every nurse note is a draft until a human approves it; the patient-facing text is checked by code. Real wound photos come only from public, de-identified research datasets [6-8], not redistributed; only the CC BY 4.0 set appears (a figure, one app sample). The weights are for non-commercial use (part of the training data is CC BY-NC). The author's own photos stay out of the repo. Private S3, least-privilege role, 14-day logs, daily cap.

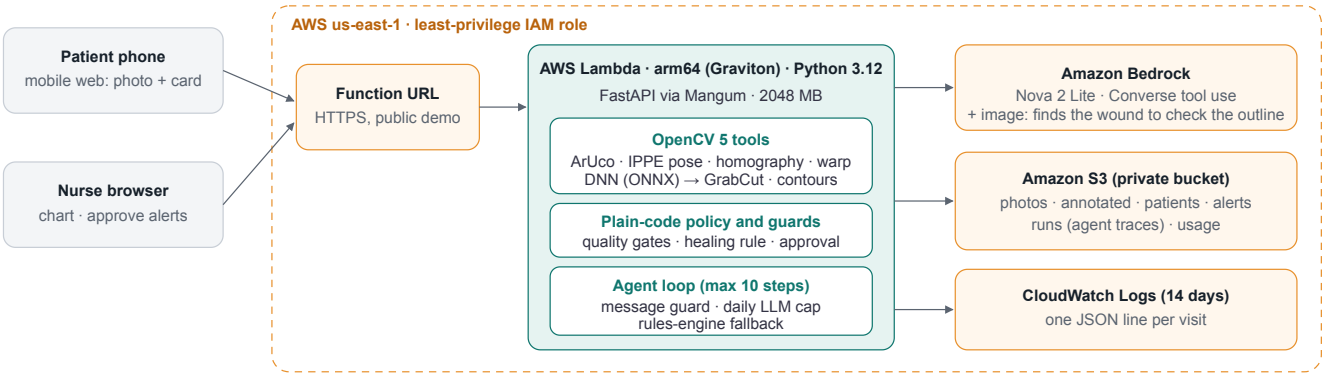
9. References

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2. Sheehan P, et al. Percent change in wound area of diabetic foot ulcers over a 4-week period is a robust predictor of complete healing in a 12-week prospective trial. Diabetes Care 2003;26(6):1879-82.
3. Carter MJ, et al. Chronic wound prevalence and the associated cost of treatment in Medicare beneficiaries: changes between 2014 and 2019. J Med Econ 2023;26(1):894-901.
4. ISO/IEC 7810: ID-1 card format, 85.60 × 53.98 mm.
5. Amazon Nova 2 Lite on Bedrock, about $0.33 / $2.75 per million input / output tokens (third-party trackers).
6. Wang C, et al. Fully automatic wound segmentation with deep convolutional neural networks. Sci Rep 2020;10:21897; and the Foot Ulcer Segmentation Challenge (FUSeg), MICCAI 2021. Photos: AZH Wound and Vascular Center, shared by the authors with the clinic's permission.
7. Sánchez-Jiménez D, et al. CO2Wounds-V2: Extended chronic wounds dataset from leprosy patients. 2024. Mendeley Data, doi:10.17632/s2w7rjwz49.2 (CC BY-NC 3.0).
8. Islam MM, et al. Lower Limb and Feet Wound Image Dataset for Medical Analysis. 2025. Mendeley Data, doi:10.17632/hsj38fwnvr.3 (CC BY 4.0).

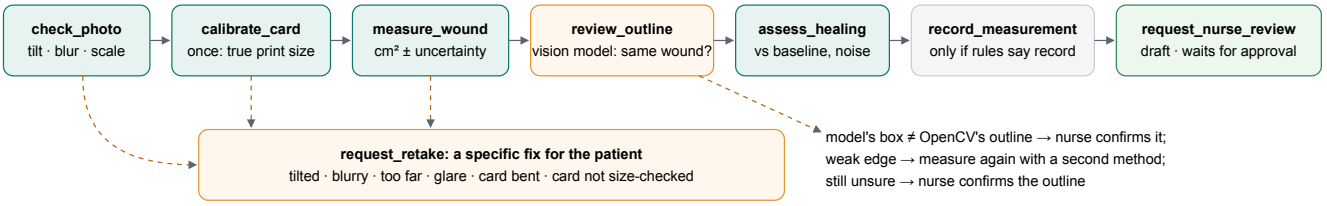
Architecture

WoundTrace: architecture on AWS

OpenCV 5 measures. An agent on Amazon Bedrock decides the next step. Plain code gates every action. A nurse approves every alert.



One visit: perceive → decide → act



Solid: normal path. Dashed: the image result changes the plan. The LLM picks tools and words messages; it cannot override a gate.

Real wound examples (CC BY 4.0 dataset, never used in training)



90th percentile: Dice 0.84
dataset mask (white), ours (green)



50th percentile: Dice 0.36
dataset mask (white), ours (green)



10th percentile: Dice 0.00
dataset mask (white), ours (green)



healthy foot (held out)
no outline drawn: correct

Photos: Lower Limb and Feet Wound Image Dataset (Islam et al., Mendeley Data, CC BY 4.0). Never used to train our model. Its masks include skin around the wound.